Thermodynamics Assignment 2

1. Show that the total work produced by a Carnot cycle per mole of the ideal gas depends solely on the temperature difference between the hot and cold reservoir
2. Show that violation of Clausius statement of 2nd law would also imply the violation of Kelvin-Planck statement.
3. Can a system undergo a spontaneous process at const. volume, such that its entropy remains unchanged? If so, how can such a process be consistent with the 2nd law that requires $\Delta S_{Univ} > 0$?